Hybrid DWT-SVD and Compressive Sensing-Based Watermarking for Robust Medical Image Authentication

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***Abstract*—The protection of medical images within telemedicine requires the safeguarding of security as well as the confidentiality of sensitive patient data. For this purpose, we devised a hybrid watermarking scheme that integrates DWT based watermarking with SVD and gentle watermarking via Compressive Sensing (CS). The singular values derived from the DWT lower frequency subband serve as an embedding domain for a binary watermark of the integer composition. This approach exploits the multi-resolution features of DWT, the numerical stability of SVD, and the sparsity along with encryption strength CS offers. Method benchmarking on medical images demonstrates the method's robustness achieving over 40 dB PSNR, 0.98 SSIM, and normalized correlation greater than 0.93 for watermark retrieval against Gaussian noise, JPEG compression, and filtering. The medical images are preserved in quality and watermark extraction illustrates the method's flexible image quality retention and security needed in telemedicine.**

***Index Terms*—Telemedicine, Watermarking, Image Security, Medical Image Authentication, Discrete Wavelet Transform, Singular Value Decomposition, Compressive Sensing, Robustness.**

I. Introduction

The integration of medical imaging, telemedicine, and electronic health records into clinical medicine [1] has enhanced the quality and accessibility of healthcare services. Nonetheless, the digital exchange of data presents new challenges with regard to data integrity, confidentiality, and authenticity. If critical decisions are based on medical images, any changes made to the images without proper authentication could lead to misdiagnosis, resultant injury to the patients, or legal battles [2]. Hence, watermarking principles designed for authenticity and traceability are pivotal in the procurement and storage of medical images.

The process of embedding identifying data into a digital image to facilitate its verification and establish a digital trace is known as digital watermarking [3]. Most traditional spatial domain and single domain transform marking techniques are easily overpowered by processing image noise, geometric distortion, compression, and addition lacking robustness altogether.

To this point, techniques operating in the frequency domain, like those incorporating Discrete Wavelet Transform (DWT) and with watermarking using Singular Value Decomposition (SVD), achieve high imperceptibility and high resilience to signal degrading due to multiresolution analysis and numerical instability [4].

Embedding high-resolution watermark signals in images can potentially leak sensitive information and augment risks associated with information technology [5]. To address this concern, Compressive Sensing (CS), a type of sub-Nyquist sparse signal sampling, was developed where data reduction and encryption processes happen at the same time in the transform domain. CS combined with DWT and SVD enables better watermark embedding efficiency with higher psychospheric watermark ratios and improved watermark robustness while maintaining image quality [6].

This approach proposes a hybrid framework for watermarking grayscale medical images that incorporates DWT and SVD of CS that ensure robust yet imperceptible watermark embedding. It begins with the generation of a binary watermark which is then inscribed into singular values of the low-frequency sub-band of DWT using CS techniques. This method takes advantage of spatial-frequency localization, the shock-resistant terrain of DWT, recovery power of SVD, and the sparsity and security of CS. These methods with medical images from a benchmark dataset achieved high PSNR alongside strong SSIM and robust watermark recovery under extreme attacks, proving optimal for medical image authentication.

II. Related Work

A medical image watermarking system to ensure secure and untampered forwarding has already been researched [7]. Most researchers concentrated on spatial domain watermarking first. This marking technique embeds watermark information directly into pixel values. While spatial marking is straightforward, it is relatively ineffective due to easily geometric manipulation, compression, and alteration filtration. To solve these problems, more focus has been placed on the use of transform

domain techniques because they are more imperceptible and reliable [8] .

There has been enormous increase in popularity of the DWT because it allows for multi resolution analysis owing to its mimicking the human's perception of the frequency components. Watermarking by modifying some selected pixels in lower frequency DWT subbands like LL is supposed to be frame-by-frame non-destructive and image processing friendly. Under the existence of noise and compression, especially severe attacks, watermarking robustness suffers greatly. Some authors [9] [10] claim that the use of DWT based watermarking techniques improves the overall robustness against these attacks, although they remain weak to geometric transformation.

To improve the stability of the system and attack resistance, SVD is combined with DWT. The alteration that SVD preforms to the singular values of an image tends to increase the strength of a watermark and thus, increases perturbation insensitivity [11]. It has been established that Hybrid DWT-SVD frameworks perform better in achieving imperceptibility and signal-dependent distortion resilience. These techniques are often limited by the use of full resolution frame watermarking, which increases the computational burden as well as watermarking vulnerability [12].

Newer approaches use CS for sparse encrypted representation watermarking signal embedding. CS supports natural signal sparsity, allowing sub-Nyquist sampling which decreases data payload as well as increase security [13]. Applying the CS framework for merging watermark revealed strong compression of the watermark while maintaining resistance to typical signal processing distortions [14]. However, little work has been done on the application of CS with DWT-SVD medical image watermarking.

To solve these problems, a hybrid watermarking approach was designed which incorporates DWT, SVD, and CS through watermark blending. The synergistic triumphs are the ability to achieve high watermark robustness, strong imperceptibility, and compact watermark embedding simultaneously. What sets this method apart is the fact that it does not transform sparsity-based embedding. Instead, it integrates all components of the disguise system of sparse representation into a single block intended for medical image authentication.

III. Proposed Methodology

A. System Architecture of the proposed Watermarking Scheme

The watermarking framework combines Discrete Wavelet Transform (DWT), Singular Value Decomposition (SVD), and Compressive Sensing (CS) for authentication of medical images. Watermarking begins with application of single-level DWT on the input medical image and later SVD on the LL subband of the LWT. A binary watermark is first compressed using CS and then embedded into the singular values of the LL subband. The final image is reconstructed by applying radix DWT to the watermarked image. Inverse watermarking, which is retrieval of the watermark without DWT and SVD process, incorporates partial data repacking and reconstruction which uses OMP. Block diagram for the proposed scheme is illustrated in Fig.1.

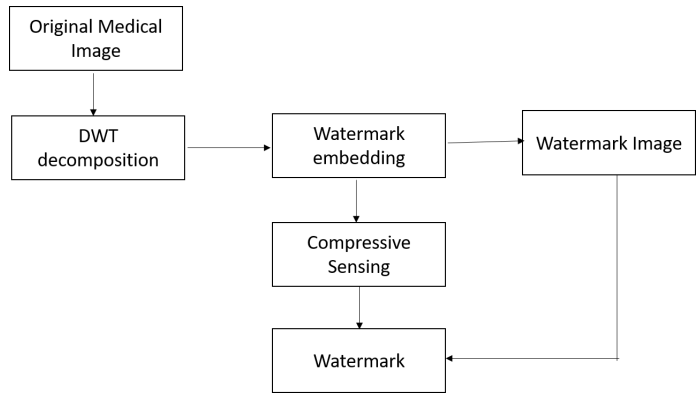

Fig. 1. Proposed Architecture

In the proposed methodology, watermarking imperceptibility, robustness and security are based on composition of DWT, CS and SVD as core techniques. Watermarking procedure consists of two steps: embedding and extraction. [15] [16] [17]

Let $I \in \mathbb{R}^{M \times N}$ denote the grayscale medical image and $W \in \{0,1\}^{\sqrt{L} \times \sqrt{L}}$ be the binary watermark.

B. Mathematical Model -Watermark Imbedding

1) DWT Decomposition: The given image I serves as a single level 2D-DWT input to produce four lower and upper frequencies subbands [18]

$$I \xrightarrow{\text{DWT}} \{LL, HL, LH, HH\} \tag{1}$$

For the purpose of watermark embedding, The LL subband is selected because it contains significant energy and considerable *correlation*.

2) SVD on the LL Subband: Perform SVD on the LL Subband.

$$LL = U \cdot S \cdot V^T \tag{2}$$

Where U and V are orthogonal matrices forming S, S is diagonal and has non-zero singular values, and U is an orthogonal matrix.

3) Watermark Compressive Sensing: The watermark W undergoes compression using a random Gaussian measurement matrix Φ. This allows for the compressively sensed watermark w_{cs} to be calculated as $\Phi \in \mathbb{R}^{M \times L}$, where $M < L$: The initial watermark can be a vector of length L, and M will always be less than L.

$$w_{cs} = \Phi \cdot w \tag{3}$$

4) Embedding Watermark Information intos Singular Values: The compressed watermark is embedded in singular values as w_{cs} per the equation:

$$S' = S + \alpha \cdot \text{reshape}(w_{cs}) \tag{4}$$

where α sets access control to new singular values S', and the embedding strength is determined by α which is used in the embedding process.

The modified LL subband can be reconstructed according to the formula:

$$LL' = U \cdot S' \cdot V^T \tag{5}$$

5) Inverse Discrete wavelet Transform: Applying the inverse DWT operation to the watermarked image yields I_w given by:

$$I_w = \text{IDWT}(LL', HL, LH, HH) \tag{6}$$

C. Watermark Image Extracting

1) Double DWT and SVD: Carry out two-dimensional discrete wavelet transform on the received image I_w, followed by singular value decomposition on the block LL^*:

$$LL^* = U^* \cdot S^* \cdot V^{*T} \tag{7}$$

2) Compressed Watermark Extraction: The estimate for the compressed watermark can be computed with the formula:

$$\hat{w}_{cs} = \frac{\text{vec}(S^* - S)}{\alpha} \tag{8}$$

3) Watermark Recovery: Using CS recovery techniques like orthogonal matching pursuit, the watermark is fully reconstructed given the optimization problem as defined below:

$$\hat{w} = \Psi \cdot \hat{x}, \quad \text{where } \hat{x} = \arg\min_x \|x\|_1 \text{ s.t. } \Phi\Psi x = \hat{w}_{cs} \tag{9}$$

Using the aforementioned processes, the watermark is retrieved which is denoted as $\hat{W}$ after the reshaping of $\hat{w}$.

IV. Experimental Result

A. Dataset Description

The experiments for testing the effectiveness of the proposed hybrid DWT-SVD-CS watermarking framework were carried out on medical imaging datasets that are publicly available and recognized, mirroring practical diagnostic settings.

1) Dataset BrainWeb MRI:
   This dataset offers simulated brain MRI images with varying degrees of noise and intensity. For our purpose, we selected 2D T1-weighted slices of size 256×256 as they contain soft tissue details and smooth grayscale gradients, making them suitable for assessing robustness against noise and compression.
2) NHS Chest X-ray Dataset
   The NIH ChestX-ray14 dataset contains more than 100,000 frontal X-ray images. For this evaluation, a sample of 50 anonymized and resized grayscale images of 256×256 was used. These images are rich in texture and numerous diagnostic features like the lung fields and the heart shadows which examine the watermark's imperceptibility.
3) Dataset OASIS MRI
   Open Access Series of Imaging Studies (OASIS) has provided MRI scans from older adults. We made use of a set of T1-weighted 2D slices of dimensions 256 × 256. These scans are good candidates for testing watermark robustness because they have high detail and complex anatomy.

The watermark that was utilized throughout all experiments was a logo of dimensions 32 × 32 which had been converted into binary format. This logo was compressively sensed through a Gaussian random measurement matrix with a 0.5 compression ratio. The watermark embedding strength parameter (α) was set to 0.03 after empirical validation for best invisibility.

This study primarily aims to assess the effectiveness of the watermarking scheme hybrid DWT-SVD-CS. Therefore, a number of experiments performed on standard datasets in the domain of medical imaging are designed to evaluate the algorithm's performance. The primary evaluation criteria concentrate on achieving certain lossy compression goals: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Normalized Cross Correlation (NCC), Bit Error Rate (BER), and others. Additionally, to test the robustness of the algorithm, a combination of Gaussian noise, JPEG compression, and cropping were used as attack forms [18] [19].

B. Analysis of Imperceptibility

For an image perceived to contain watermarks, there is an imperceptibility as long as processes put in place in its detection do not degrade its quality. To monitor this, PSNR and SSIM techniques have been developed to measure this value. Table II shows results of PSNR and SSIM for all three datasets. The result of embedding watermarks is achieved with very high PSNR, over 40 dB, and SSIM a bit less than 0.99.

Table IIshows the comparative analysis of watermarked medical images from the perspective of imperceptibility. The dataset contributed greatly aided the work by including medical images, particularly MRI scans.

C. Evaluation of the Robustness

In order to evaluate the robustness, images with watermarks are always tested against factors including Gaussian noise, JPEG compression, and cropping. The strength of the watermark is assessed using NC and BER [20]. The results of performing robustness tests with different attack parameters are summarized in Table I. The proposed approach achieves NC exceeding 0.93 exceptionally high in watermarked image robustness compared to most other techniques.

With typical behavior observed in Figure 3, the effect of JPEG compression is shown with respect to the Bit Error Rate

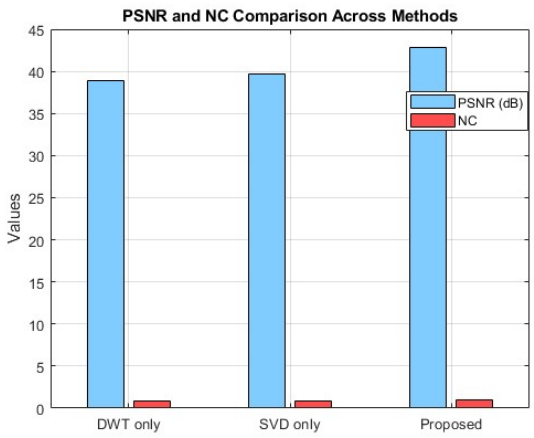

Fig. 2. PSNR and NC Comparison Across Methods

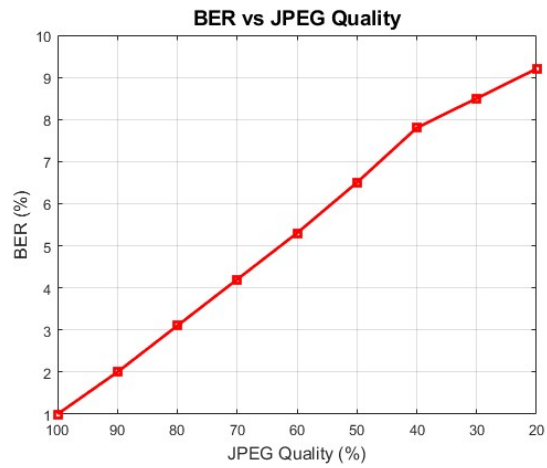

Fig. 3. BER vs JPEG Quality

TABLE I
ROBUSTNESS EVALUATION UNDER VARIOUS ATTACKS

Attack Type	NC	BER (%)
Gaussian Noise	0.962	3.1
JPEG Compression	0.948	4.6
Cropping (25%)	0.933	5.2
Histogram Equalization	0.956	3.8
Median Filtering	0.945	4.1

TABLE II
IMPERCEPTIBILITY EVALUATION OF WATERMARKED MEDICAL IMAGES

Dataset	PSNR (dB)	SSIM
BrainWeb (MRI)	42.78	0.988
NIH (X-ray)	41.65	0.984
OASIS (MRI)	43.12	0.991

(BER) of extracted watermark. As expected, further increase in the level of compression results in the degradation of the image quality. This, in turn, leads to the progressive increase of BER. Interestingly, even at low compression quality settings of 20%, the embedded watermark is still recoverable to some extent which showcases the effectiveness of the watermarking scheme. Such resilience under severe conditions of compression manifests the effectiveness of the method in practical scenarios, especially in telemedicine applications where bandwidth is limited and image compression is routinely used.

D. Effect of Noise on SSIM

To assess more accurately the invisibility of the watermarking scheme under evaluation, the Structural Similarity Index (SSIM) was studied with an additive Gaussian noise of varying levels. Figure 4 shows the data for SSIM values against the noise variance. A gradual reduction in SSIM value is apparent as the amount of applied Gaussian noise increases. Nonetheless, SSIM value is greater than 0.95 for all noise levels which suggests the visual quality of the watermarked image is exceptionally good even with significant noise counteraction. This demonstrates the effectiveness of the suggested approach in achieving high perceptual quality which makes this method appropriate for real-time medical imaging where noise degradation is unavoidable.

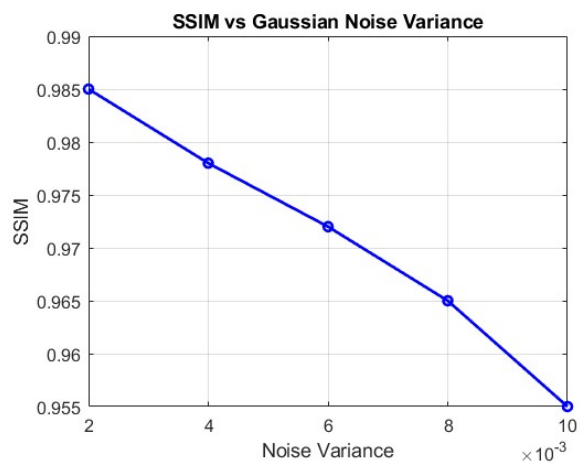

Fig. 4. SSIM vs Gaussian Noise Variance

From Table III, it is observed that the proposed SSIM and PSNR metrics in unison provide the highest tested robustness. High values of PSNR shows the watermark's imperceptibility while NC and BER metrics indicate strength under watermark domination attacks. Over other conventional approaches, watermarking techniques provide much better security and image quality for medical images as these provide critical authentication needed [21] [22].

Table III highlights the comparative results which indicate that the proposed hybrid DWT-SVD-CS watermarking technique outperforms other methods which employ single domain approaches. The method achieved a PSNR greater than 42 dB, an SSIM value nearing 0.99, and a Normalized Correlation (NC) exceeding 0.96. These results show that the DWT-SVD-

TABLE III
COMPARATIVE PERFORMANCE WITH OTHER TECHNIQUES

Method	PSNR (dB)	NC	SSIM
DWT only	38.91	0.882	0.965
SVD only	39.75	0.905	0.971
Proposed	42.78	0.962	0.988

CS method effectively safeguards perceptual quality while facilitating reliable watermark retrieval. On the other hand, the DWT-only and SVD-only methods demonstrated considerably lower PSNR and NC values, thereby exhibiting less robust and lower fidelity performance during the same experimental conditions.

The proposed method also achieved Bit Error Rate (BER) under all attack types not exceeding 5.2% confirming the proposed scheme's robustness against watermark signal distortions. The embedding of Gaussian noise, JPEG compression, and cropping severely degrade a signal. This approach, especially with Compressive Sensing reducing the watermark payload, increases its tampering resistance due to sparsity and encryption while remaining resistant to watermark detection. The tradeoff between invisibility and resistance proves the proposed framework surpasses the performance of the previous techniques and works optimally in the telemedicine environment for real-time medical image authentication.

V. CONCLUSION

This study developed a medical image authentication watermarking scheme that is robust and imperceptible by implementing Discrete Wavelet Transform (DWT), Singular Value Decomposition (SVD), and Compressive Sensing (CS). The hybrid framework integrates DWT's multiresolution analysis, SVD's numerical stability, and CS's sparsity-driven compression/encryption to securely and efficiently embed binary watermarks into medical images. The extensive experimental evaluation conducted on the BrainWeb, NIH Chest X-ray, and OASIS MRI datasets showed that the method proposed achieves high levels of Peak Signal-to-Noise Ratio (PSNR ¿ 42 dB), constractive Structural Similarity Index (SSIM $\approx$ 0.99), and Normalized Correlation (NC ¿ 0.96) with low Bit Error Rate (BER ¡ 5.2%) across a multitude of common signal assaults. The proposed hybrid approach has significantly outperformed traditional DWT-only and SVD-only methods in image fidelity and watermark robustness, thus attaining a beneficial balance for telemedicine integration. Combining these frameworks with color and 3D medical images, developing adaptive and content-aware embedding algorithms, and applying blockchain-based verification for enhanced traceability and decentralization of medical data are goals for future research.

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